Q09 Comprehension Check

Question: A colleague proposes the following pretraining data plan: "We have 100B tokens of curated text. Chinchilla says scale data, so let's train on each document *40 epochs* to get 4T effective training tokens — repetition is free." Identify the error, cite the relevant empirical result, and state what the evidence actually supports.

Answer: The error is treating repeated tokens as equivalent to fresh tokens. Muennighoff et al. (2023) varied repetition and compute across 400+ runs and found that with constrained data, up to about **4 epochs** of repeated data yields loss changes that are negligible relative to unique data; beyond that, the value of added compute decays toward zero. So 40 epochs gives nowhere near a 40× increase in effective tokens — after ∼4 epochs, returns have largely vanished. The evidence supports: (1) repeating a clean corpus up to ∼4 epochs as a reasonable way to spend fixed compute (with the data-constrained optimum shifting toward smaller models trained more epochs, not Chinchilla's equal parameter/data scaling); and (2) complementing scarce text with other data (e.g., code, which Muennighoff et al. find can roughly double effective tokens even for natural-language evaluation) rather than hammering the same tokens.